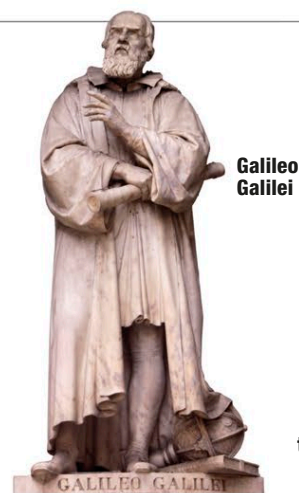
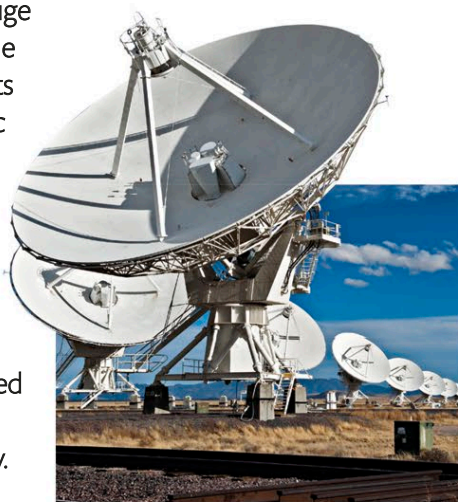


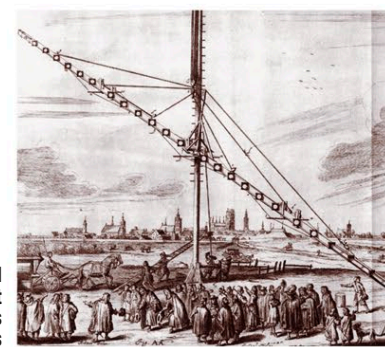
THE HISTORY OF THE TELESCOPE

TELESCOPES ARE THE VITAL TOOLS OF AN ASTRONOMER'S TRADE, IMPROVING ON HUMAN EYESIGHT AND ALLOWING IMAGES AND DATA TO BE PROCESSED IN DIFFERENT WAYS, AND RECORDED FOR POSTERITY.

Dutch lensmaker Hans Lippershey is usually credited with inventing the telescope around 1608, but it was Italian physicist Galileo Galilei who first turned it toward the sky. Since then, telescope technology has gone through huge advances—the introduction of the mirrored reflector design, mounts that can keep a telescope in sync with the movement of the stars, spectroscopy to analyze the chemical fingerprints of starlight, and photography to keep a permanent record of observations. More recently, computer control and space-based observatories have helped push the limits of telescope technology.



Galileo Galilei



Aerial telescope built by Johannes Hevelius

1609

Galileo's telescope

Galileo's first lens-based telescope produced an image multiplied by a factor of three, but later designs improved rapidly. He used his instruments to make discoveries including mountains on the Moon, spots on the Sun, and countless stars invisible to the naked eye.

1673

Aerial telescopes

One way of improving the magnification of telescopes was to use larger lenses separated by a greater distance. In the mid-17th century this led to enormous aerial telescopes with lenses suspended on open frames up to 150 ft (31 m) long.

High-altitude twin Keck telescopes on Mauna Kea, Hawaii



Very Large Array, New Mexico

1980

Telescope arrays

A technique called interferometry allows for the combination of signals from several telescopes to mimic the resolving power of a single, impossibly large instrument. The technique was pioneered for use with the long-wavelength radio waves at the Very Large Array.

1970

Orbiting observatories

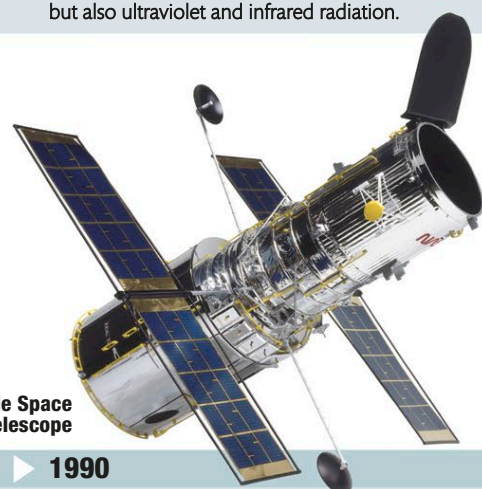
The launch of the first X-ray astronomy satellite, Uhuru, heralded a new age of space-based astronomy, studying the Universe at wavelengths that are blocked by Earth's atmosphere. These included not only X-rays but also ultraviolet and infrared radiation.

1949

Mountaintop telescopes

Astronomers had long recognized that observing from high altitudes helped reduce the atmospheric turbulence affecting starlight, but it was only in the second half of the 20th century that observatories on remote mountaintops became practical.

Hubble Space Telescope



1980s

Segmented-mirror telescopes

The weight of traditional mirrors brought the growth of telescopes to a halt in the mid-20th century, but from the 1980s onward, engineering breakthroughs allowed telescopes to reach even bigger sizes. Key to this was the ability to align honeycomb-like mirror segments to mimic a single reflecting surface.

1990

Hubble Space Telescope

The idea of placing a large optical telescope above Earth's atmosphere, where it would experience perfect observing conditions, was suggested as early as 1946. The ability to repair and upgrade the Hubble Space Telescope in orbit has kept it functional for more than a quarter century. While modest in size compared to today's ground-based giants, Hubble's location allows it to deliver both stunning images and revolutionary scientific discoveries.